

Research Progress Update – Dynamic Qubit Allocation Framework & Validation Results

Piter Garcia (RIT Student) <pzg8794@rit.edu>
 Reply-To: pzg8794@rit.edu
 To: Daniel Krutz <dxkvse@rit.edu>

Tue, Oct 21, 2025 at 11:55 AM

Dan,

I am so sorry-I could have sworn your name was Liam, I don't even know anyone close with that name. The last person I remember with that name was my first boss, I have no clue how I mixed them up; that is a signal of how sleep deprived my brain is right now 😊 .

Thank you so much for your understanding, I really appreciate it. No worries, health has always been a priority to me, which is the main reason why I am back here.

Yes, I have completed all of the steps, but to be honest, I need to go back to them, review them once more, and make any changes if needed. Something I was planning to do today before referencing the document/template you provided me with. I will make be emailing you soon with a layout of the documentation, which I have already but I will make sure that it is concise and clear for you.

Wednesday at 1 pm works great for me, it will give me plenty time to continue recovering.

Thank you for your understanding, and for being very considerate in a time like this, I appreciate it very much.

Sincerely,
 Piter

On Tue, Oct 21, 2025 at 8:14AM Daniel Krutz <dxkvse@rit.edu> wrote:

Hi Piter, thanks for the report. First and foremost I hope that your dad is doing ok. Make sure to focus on family first. Also, my name isn't Liam :-p

1. Have you completed a literature review comparing this proposed process against existing works? Also, have you created a set of research questions and evaluation plan? (You've shared a lot of docs with me, so I want to make sure that I am not missing anything).

How does Wednesday at 1 work for a Zoom?

Thanks
 -Dan

On Oct 21, 2025, at 4:17 AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hi Liam,

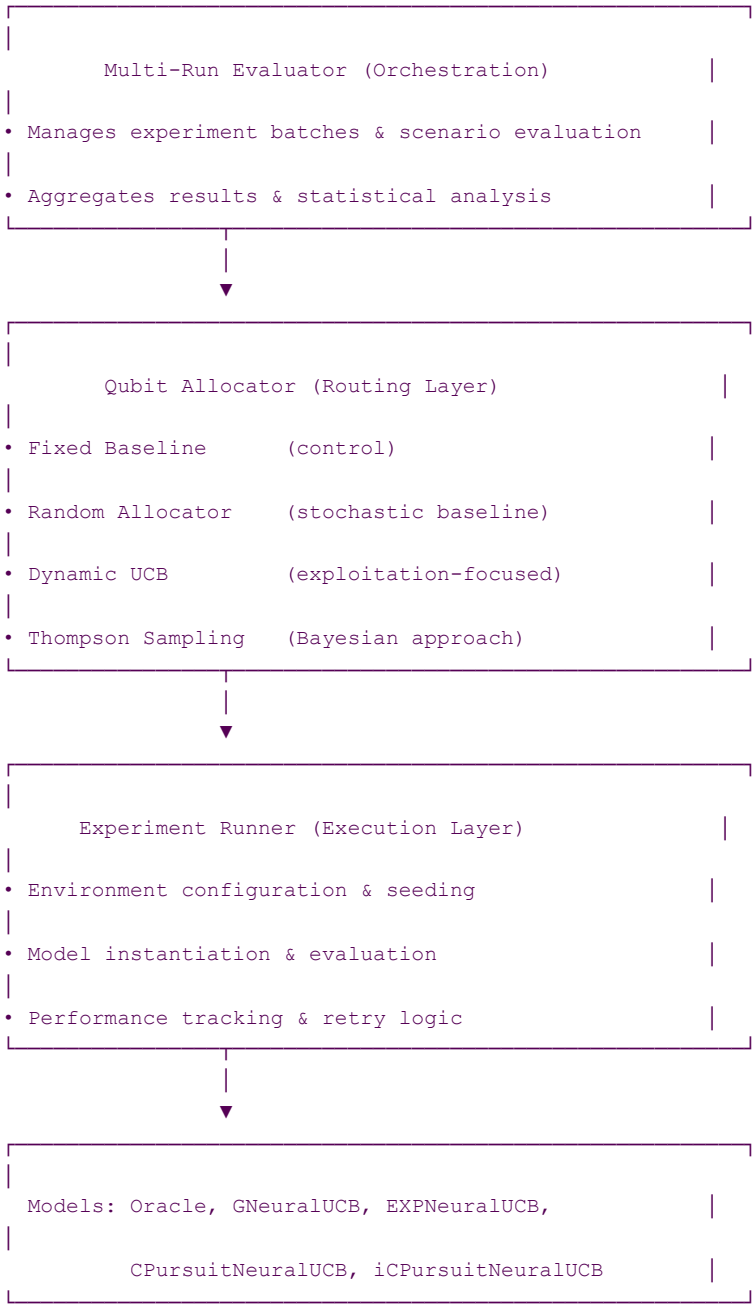
I hope this email finds you well.

I wanted to provide you with an update on my research progress, particularly given the challenging week I've had with my father's hospitalization through Friday. While balancing family obligations and catching up on coursework, I've been making exciting strides on the quantum routing research that I'm eager to share with you.

Framework Development:
Major Milestone Achieved
 I'm thrilled to report that we've completed development and initial validation of a **robust,**

production-ready evaluation framework for quantum entanglement routing. This framework represents a significant evolution from our previous implementations and addresses all the stability and reproducibility issues encountered in some experiments.

Framework Architecture
(Simplified View)



Key Framework Features

-

Modular routing strategies:
Easy integration of new allocation algorithms

-

Deterministic reproducibility:
Fixed seeds and isolated experiment states

-

Automatic retry & validation:

Built-in threshold detection and recovery

-

Comprehensive logging:

Full traceability from allocation → execution → results

Preliminary Results: Qubit

Allocation Strategy Comparison

I've completed initial

comparative experiments testing **four**

qubit allocation strategies

across stochastic quantum network conditions. Here's a high-level summary:

Thompson Sampling Allocator

(Bayesian Approach)

-

Best performing

in 4000-frame experiments: **94.97%**

efficiency (iCPursuitNeuralUCB)

-

Consistent 93-95% efficiency

across 6000-frame tests

-

Demonstrates strong exploitation-exploration

balance

Random Allocator (Stochastic

Baseline)

-

Surprisingly competitive:

75.44% avg efficiency at 4000 frames

-

Degrades at scale: 69-74%

at 8000 frames

-

Validates the need for

sophisticated routing (random ≠ optimal)

Dynamic UCB Allocator (Exploitation-Focused)

-

Strong mid-scale performance:
94.68-94.92% at 6000 frames

-

Some instability at 8000
frames (CPursuitNeuralUCB failure case observed)

-

Shows promise but requires
robustness tuning

Fixed Baseline (Control - INCOMPLETE)

-

Still running final 8000-frame
experiment

-

Early results show an 81-95%
efficiency range

-

Will provide a complete
comparison once finished

The framework successfully isolates allocation strategy impact from model performance—something my previous implementations struggled with due to entangled parameter dependencies. Thompson Sampling consistently outperforms random allocation by 15-25 percentage points, confirming that intelligent routing provides substantial gains. You can access the latest notebooks, with this and more results, by accessing this [link](#).

Next Steps & Validation

Plan

With this stable framework now operational, my immediate priorities are:

- 1.

Complete & document new experiments (awaiting final run completion)

- 2.

Validate all previous framework results using this robust implementation to resolve inconsistencies

3.

Conduct ablation studies

on epsilon-greedy variants and annealing schedules

4.

Documentation,

which I am starting
tomorrow, referencing the template you provided.

This framework provides
the systematic validation tool we need to confidently publish our findings and understand the true
performance characteristics of each model developed and observed.

Despite the challenging
circumstances with my father's health this week, working on this research has been a source of focus and
motivation. The breakthrough in framework stability came after debugging through the night, and seeing
these clean, reproducible results has been incredibly
rewarding.

I'm excited to discuss
these findings with you and get your thoughts on prioritizing the validation experiments. I cannot meet on
Tuesday morning because I have to take my dad to the doctor for multiple appointments he has in the
morning. Please let me know when you have time to
meet—Tuesday (today) afternoon or evening works, or Wednesday/Thursday after 10 AM, since I'm
trying to catch up on sleep. I'd love to walk you through the full experimental design and preliminary
visualizations.

Thank you for your continued
support and understanding during this difficult period.

Best,
Piter